

Saklovir 250 mg Powder for Infusion
Acyclovir

Description

White to off white Lyophilized powder

Description after reconstitution
A Clear colorless Solution.

Description after dilution.
A Clear colorless Solution.

When diluted in accordance with the recommended schedules, Acyclovir for Injection 250 mg is known to be compatible with the following infusion fluids and stable for up to 6 hours at room temperature (15°C to 25°C):

Compound Sodium Lactate Intravenous Infusion (Hartmann's Solution)
Sodium Chloride Intravenous Infusion
Glucose Intravenous Infusion

Composition

Acyclovir 250.0 mg

Mode of action

Aciclovir is a synthetic purinenucleoside analogue with in vitro and in vivo inhibitory activity against human herpes viruses, including herpes simplex virus (HSV) types 1 and 2, varicella zoster virus (VZV), Epstein Barr virus (EBV) and cytomegalovirus (CMV). In cell culture, aciclovir has the greatest antiviral activity against HSV-1, followed (in decreasing order of potency) by HSV-2, VZV, EBV and CMV. The inhibitory activity of aciclovir for HSV-1, HSV-2, VZV, EBV and CMV is highly selective. The enzyme thymidine kinase (TK) of normal, non-infected cells does not use aciclovir effectively as a substrate, hence toxicity to mammalian host cells is low; however, TK encoded by HSV, VZV and EBV converts aciclovir to aciclovir monophosphate, a nucleoside analogue, which is further converted to the diphosphate and finally to the triphosphate by cellular enzymes. Aciclovir triphosphate interferes with the viral DNA polymerase and inhibits viral DNA replication with resultant chain termination following its incorporation into the viral DNA.

Summary of Pharmacodynamics and Pharmacokinetics

Pharmacodynamic Effects

Prolonged or repeated courses of aciclovir in severely immune-compromised individuals may result in the selection of virus strains with reduced sensitivity, which may not respond to continued aciclovir treatment. Most of the clinical isolates with reduced sensitivity have been relatively deficient in viral TK however, strains with altered viral TK or DNA polymerase have also been reported. In vitro exposure of HSV isolates to aciclovir can also lead to the emergence of less sensitive strains. The relationship between the in-vitro determined sensitivity of HSV isolates and clinical response to aciclovir therapy is not clear. All patients should be cautioned to ensure they avoid the potential of virus transmission, particularly when active lesions are present.

Pharmacokinetic Properties

Absorption
Not Applicable.

Distribution

The mean volume of distribution of 26 L indicates that aciclovir is distributed within total body water. Apparent values after oral administration (Vd/F) ranged from 2.3 to 17.8 L/kg. As plasma protein binding is relatively low (9 to 33%), drug interactions involving binding site displacement are not anticipated. Cerebrospinal fluid levels are approximately 50% of corresponding plasma levels at steady-state.

Metabolism

Aciclovir is predominantly excreted unchanged by the kidney. The only known urinary metabolite is 9-[(carboxymethoxy) methyl]guanine, and accounts for 10-15% of the dose excreted in the urine.

Elimination

Mean systemic exposure (AUC_{0-∞}) to aciclovir ranges between 1.9 and 2.2 microgram*h/mL after a 200 mg dose. In adults the terminal plasma half-life after oral administration has been shown to vary between 2.8 and 4.1 hours. Renal clearance of aciclovir (CL_R= 14.3 L/h) is substantially greater than creatinine clearance, indicating that tubular secretion, in addition to glomerular filtration, contributes to the renal elimination of the drug. The half-life and total clearance of aciclovir are dependent on renal function. Therefore, dosage adjustment is recommended for renally impaired patients. In neonates (0 to 3 months of age) treated with doses of 10mg/kg administered by infusion over a one-hour period every 8 hours the terminal plasma half-life was 3.8 hours.

Special Patient Populations

In patients with chronic renal failure the mean terminal half-life was found to be 19.5 hours. The mean aciclovir half-life during haemodialysis was 5.7 hours. Plasma aciclovir levels dropped approximately 60% during dialysis. In the elderly total body clearance falls with increasing age, associated with decreases in creatinine clearance, although there is little change in the terminal plasma half-life.

Indications

Saklovir for infusion is indicated for the treatment of herpes simplex infections.
Saklovir for infusion is indicated for the prophylaxis of herpes simplex infections in immune-compromised patients.
Saklovir for infusion is indicated in the treatment of varicella zoster infections.
Saklovir for infusion is indicated for the treatment of herpes simplex infections in the neonate.

Adverse reactions/side effects

Blood and lymphatic system disorders:
Uncommon: decreases in haematological indices (anaemia, thrombocytopenia, leukopenia).
Immune system disorders:
Very rare: anaphylaxis.

Psychiatric and nervous system disorders:

Very rare: headache, dizziness, agitation, confusion, tremor, ataxia, dysarthria, hallucinations, psychotic symptoms, convulsions, somnolence, encephalopathy, coma.

The above events are generally reversible and usually reported in patients with renal impairment or with other predisposing factors.

Vascular disorders:

Common: phlebitis.

Respiratory, thoracic and mediastinal disorders:
Very rare: dyspnoea.

Gastrointestinal disorders:
Common: nausea, vomiting.
Very rare: diarrhoea, abdominal pain.

Hepato-biliary disorders:
Common: reversible increases in liver-related enzymes.
Very rare: reversible increases in bilirubin, jaundice, hepatitis.

Skin and subcutaneous tissue disorders:
Common: pruritus, urticaria, rashes (including photosensitivity).
Very rare: angioedema.

Renal and urinary disorders:
Common: increases in blood urea and creatinine.
Rapid increases in blood urea and creatinine levels are believed to be related to the peak plasma levels and the state of hydration of the patient. To avoid this effect the drug should not be given as an intravenous bolus injection but by slow infusion over a one-hour period.
Very rare: renal impairment, acute renal failure and renal pain.
Adequate hydration should be maintained. Renal impairment usually responds rapidly to rehydration of the patient and/or dosage reduction or withdrawal of the drug. Progression to acute renal failure however, can occur in exceptional cases.
Renal pain may be associated with renal failure and crystalluria.

General disorders and administration site conditions:
Very rare: fatigue, fever, local inflammatory reactions

Severe local inflammatory reactions sometimes leading to breakdown of the skin have occurred when acyclovir has been inadvertently infused into extracellular tissues.

Precaution/warnings.

In patients receiving Saklovir for infusion at higher doses (e.g. for herpes encephalitis), specific care regarding renal function should be taken, particularly when patients are dehydrated or have any renal impairment. Reconstituted Saklovir for infusion has a pH of approximately 11.0 and should not be administered by mouth. Use in patients with renal impairment and in elderly patients.

Aciclovir is eliminated by renal clearance, therefore the dose must be reduced in patients with renal impairment. Elderly patients are likely to have reduced renal function and therefore the need for dose reduction must be considered in this group of patients. Both elderly patients and patients with renal impairment are at increased risk of developing neurological side effects and should be closely monitored for evidence of these effects. In the reported cases, these reactions were generally reversible on discontinuation of treatment.

Contraindications

Hypersensitivity to acyclovir or valaciclovir, or to any of the excipients.

Dosage and directions for use

Route of administration: Slow intravenous infusion over 1 hour.

A course of treatment with Saklovir for infusion usually lasts 5 days, but this may be adjusted according to the patient's condition and response to therapy. Treatment for herpes encephalitis usually lasts 10 days. Treatment for neonatal herpes infections usually lasts 14 days for mucocutaneous (skin-eye-mouth) infections and 21 days for disseminated or central nervous system disease. The duration of prophylactic administration of Saklovir for infusion is determined by the duration of the period at risk.

Dosage in adults:

Patients with Herpes simplex (except herpes encephalitis) or Varicella zoster infections should be given Saklovir in doses of 5 mg/kg body weight every 8 hours provided renal function is not impaired (see Dosage in renal impairment).

Immunocompromised patients with Varicella zoster infections or patients with herpes encephalitis should be given Saklovir in doses of 10 mg/kg body weight every 8 hours provided renal function is not impaired (see Dosage in renal impairment).

In obese patients dosed with intravenous aciclovir based on their actual body weight, higher plasma concentrations may be obtained (see Pharmacokinetic properties). Consideration should therefore be given to dosage reduction in obese patients and especially in those with renal impairment or the elderly.

Dosage in infants and children: The dose of Saklovir for infants and children aged between 3 months and 12 years is calculated on the basis of body surface area.

Infants and children 3 months of age or older with Herpes simplex (except herpes encephalitis) or Varicella zoster infections should be given Saklovir in doses of 250 mg per square metre of body surface area every 8 hours if renal function is not impaired.

In immunocompromised children with Varicella zoster infections or children with herpes encephalitis, Saklovir should be given in doses of 500 mg per square metre body surface area every 8 hours, if renal function is not impaired.

The dosage of Saklovir in neonates and infants up to 3 months of age is calculated on the basis of body weight.

The recommended regimen for infants treated for known or suspected neonatal herpes is acyclovir 20 mg/kg body weight IV every 8 hours for 21 days for disseminated and CNS disease, or for 14 days for disease limited to the skin and mucous membranes.

Infants and children with impaired renal function require an appropriately modified dose, according to the degree of impairment (see Dosage in renal impairment).

Dosage in the elderly:

The possibility of renal impairment in the elderly must be considered and the dosage should be adjusted accordingly (see Dosage in renal impairment below).

Adequate hydration should be maintained.

Dosage in renal impairment:

Caution is advised when administering Saklovir to patients with impaired renal function. Adequate hydration should be maintained.
Dosage adjustment for patients with renal impairment is based on creatinine clearance, in units of ml/min for adults and adolescents and in units of ml/min/1.73m² for infants and children less than 13 years of age. The following adjustments in dosage are suggested:

Dosage adjustments in adults and adolescents:

Creatinine Clearance	Dosage
25 to 50 ml/min	The dose recommended above (5 or 10 mg/kg body weight) should be given every 12 hours.
10 to 25 ml/min	The dose recommended above (5 or 10 mg/kg body weight) should be given every 24 hours.
0(anuric) to 10 ml/min	In patients receiving continuous ambulatory peritoneal dialysis (CAPD) the dose recommended above (5 or 10 mg/kg body weight) should be halved and administered every 24 hours. In patients receiving haemodialysis the dose recommended above (5 or 10 mg/kg body weight) should be halved and administered every 24 hours and after dialysis.

Dosage adjustments in infants and children:

Creatinine Clearance	Dosage
25 to 50 ml/min/1.73m ²	The dose recommended above (250 or 500 mg/m ² body surface area or 20 mg/kg body weight) should be given every 12 hours.
10 to 25 ml/min/1.73m ²	The dose recommended above (250 or 500 mg/m ² body surface area or 20 mg/kg body weight) should be given every 24 hours.
0 (anuric) to 10 ml/min/1.73m ²	In patients receiving continuous ambulatory peritoneal dialysis (CAPD) the dose recommended above (250 or 500 mg/m ² body surface area or 20 mg/kg body weight) should be halved and administered every 24 hours. In patients receiving haemodialysis the dose recommended above (250 or 500 mg/m ² body surface area or 20 mg/kg body weight) should be halved and administered every 24 hours and after dialysis

Use in pregnancy and lactation**Pregnancy**

A post-marketing aciclovir pregnancy registry has documented pregnancy outcomes in women exposed to any formulation of aciclovir. The registry findings have not shown an increase in the number of birth defects amongst aciclovir exposed subjects compared with the general population, and any birth defects showed no uniqueness or consistent pattern to suggest a common cause. The use of aciclovir should be considered only when the potential benefits outweigh the possibility of unknown risks.

Lactation

Following oral administration of 200mg acyclovir five times a day, aciclovir has been detected in breast milk at concentrations ranging from 0.6 to 4.1 times the corresponding plasma levels. These levels would potentially expose nursing infants to aciclovir dosages of up to 0.3mg/kg/day. Caution is therefore advised if aciclovir is to be administered to a nursing woman.

Drug interactions

No clinically significant interactions have been identified.

Aciclovir is eliminated primarily unchanged in the urine via active renal tubular secretion. Any drugs administered concurrently that compete with this mechanism may increase aciclovir plasma concentrations. Probenecid and cimetidine increase the AUC of aciclovir by this mechanism, and reduce aciclovir renal clearance. However no dosage adjustment is necessary because of the wide therapeutic index of aciclovir.

In patients receiving intravenous Saklovir, caution is required during concurrent administration with drugs which compete with aciclovir for elimination, because of the potential for increased plasma levels of one or both drugs or their metabolites. Increases in plasma AUCs of aciclovir and of the inactive metabolite of mycophenolate mofetil, an immunosuppressant agent used in transplant patients, have been shown when the drugs are coadministered.

Care is also required (with monitoring for changes in renal function) if administering intravenous Saklovir with drugs which affect other aspects of renal physiology (e.g. cyclosporin, tacrolimus).

Overdosage, symptoms and treatment

Overdosage of intravenous acyclovir has resulted in elevations of serum creatinine, blood urea nitrogen and subsequent renal failure. Neurological effects including confusion, hallucinations, agitation, seizures and coma have been described in association with overdosage.

Patients should be observed closely for signs of toxicity. Haemodialysis significantly enhances the removal of acyclovir from the blood and may, therefore, be considered an option in the management of overdose of this drug.

Instruction for use**Reconstitution:**

Acyclovir for Injection 250 mg should be reconstituted using the 10 ml of Water for Injections BP to provide a solution containing 25 mg aciclovir per ml:

From the calculated dose, determine the appropriate number and strength of vials to be used.

To reconstitute each vial add the recommended volume of infusion fluid and shake gently until the contents of the vial have dissolved completely.

Administration:

The required dose of Acyclovir for Injection 250 mg should be administered by slow intravenous infusion over a one-hour period.

After reconstitution Acyclovir for Injection 250 mg may be administered by a controlled-rate infusion pump.

Alternatively, the reconstituted solution may be further diluted to give an acyclovir concentration of not greater than 5 mg/ml (0.5% w/v) for administration by infusion:

Add the required volume of reconstituted solution to the chosen infusion solution, as recommended below, and shake well to ensure adequate mixing occurs.

For children and neonates, where it is advisable to keep the volume of infusion fluid to a minimum, it is recommended that dilution is on the basis of 4 ml reconstituted solution (100 mg aciclovir) added to 20 ml of infusion fluid to give an aciclovir concentration of not greater than 5 mg/ml (0.5% w/v) for administration by infusion.

For adults, it is recommended that infusion bags containing 100 ml of infusion fluid are used, even when this would give an aciclovir concentration substantially below 0.5% w/v. Thus one 100 ml infusion bag may be used for any dose between 250 mg and 500 mg aciclovir (10 and 20 ml of reconstituted solution) but a second bag must be used for doses between 500 mg and 1000 mg.

When diluted in accordance with the recommended schedules, Acyclovir for Injection 250 mg is known to be compatible with the following infusion fluids and stable for up to 6 hours store below 30°C

Compound Sodium Lactate Intravenous Infusion (Hartmann's Solution)

Sodium Chloride Intravenous Infusion

Glucose Intravenous Infusion

Acyclovir for Injection 250 mg when diluted in accordance with the above schedule will give an aciclovir concentration not greater than 0.5% w/v.

Since no antimicrobial preservative is included, reconstitution and dilution must be carried out under full aseptic conditions, immediately before use, and any unused solution discarded.

Should any visible turbidity or crystallisation appear in the solution before or during infusion, the preparation should be discarded.

Effects on Ability to Drive and Use Machines

Saklovir for infusion is generally used in an in-patient hospital population and information on ability to drive and operate machinery is not usually relevant. There have been no studies to investigate the effect of acyclovir driving performance or the ability to operate machinery.

Incompatibilities

This medicinal product must not be mixed with other than

- sodium chloride intravenous infusion BP (0.45% and 0.9% w/v);
- sodium chloride (0.18% w/v) and glucose (4% w/v) intravenous infusion BP
- sodium chloride (0.45% w/v) and glucose (2.5% w/v) intravenous infusion BP

Dosage forms and packaging available

1 box contains 1 vial of 250mg Acyclovir

1 box contains 5 vial of 250mg Acyclovir

Storage conditions

Unopened vial: 24 months. Store below 30°C. Protect from light

After reconstitution with 10 ml and 4ml Sterilized water for Injection: 6 hours. Store below 30°C. Protect from light

After dilution with intravenous infusion fluids: 6 hours. Store below 30°C. Protect from light

Shelf-life

Unopened vial: 24 months

“Protect from light”

After reconstitution with 10 ml and 4ml Sterilized water for Injection: 6 hours

After dilution with intravenous infusion fluids: 6 hours

Registration number

MAL2021XXXXXXAZ

Manufactured by

SAKAR HEALTHCARE LIMITED

Block No. 10-13, Sarkhej-Bavla Highway,

Vill: Changodar, Dist: Ahmedabad-382213,

Gujarat – INDIA.

Product registration holder

Hikmah Pharmaceuticals Sdn Bhd

A-5-10 Empire Tower

SS16/1, Subang Jaya

47500 Selangor

Date of revision of package insert

9 March 2023